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ORIGINAL ARTICLE *Embryology*

Human embryonic development after blastomere removal: a time-lapse analysis

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BACKGROUND: Blastomere biopsy of human embryos is performed for preimplantation genetic diagnosis (PGD). The impact on further development is largely unexplored, though studies on mice suggest an influence on the hatching process. The objective of this study was to evaluate the effect of blastomere biopsy on early human embryonic development using time-lapse analysis.

METHODS: Embryos from couples undergoing PGD treatment or IVF/ICSI were included. In the PGD group, 56 human embryos had one blastomere biopsied. As controls, 53 non-biopsied IVF/ICSI embryos were selected. All embryos were cultured until 5 days after fertilization in a time-lapse incubator (EmbryoScope^{TM}). Images of embryos were acquired every 20 min. Time-points of key embryonic events were registered, and development in the two groups was compared.

RESULTS: Duration of the biopsied cell-stage in the PGD group was longer than in the control group ($P < 0.001$), causing biopsied embryos to reach subsequent embryonic stages until hatching at significantly later time-points ($P_{compaction} < 0.001$; $P_{morula} < 0.001$; $P_{earlyblast} < 0.001$; $P_{fullblast} = 0.01$), but with unchanged intervals. Embryos in the PGD group started hatching at the same time-point as the control group, but had a smaller diameter ($P < 0.001$), and a thicker zona pellucida ($P < 0.001$) when hatching. Time-lapse videos revealed that in the control group, expansion of the blastocyst caused continuous thinning of zona pellucida until the blastocyst hatched, whereas in the PGD group the blastocyst hatched through the opening in zona pellucida artificially introduced prior to the biopsy.

CONCLUSIONS: We find that blastomere biopsy prolongs the biopsied cell-stage, possibly caused by a delayed compaction and alters the mechanism of hatching.

Key words: preimplantation genetic diagnosis / blastomere biopsy / time-lapse recording / human embryo development

Introduction

Preimplantation genetic diagnosis (PGD) is offered to couples carrying a well-defined genetic abnormality, making selection of unaffected, healthy embryos for transfer possible. PGD requires embryonic DNA, which can be obtained by polar body or blastomere biopsy, or by biopsy of blastocyst trophectoderm. Blastomere biopsy is currently by far the most commonly used method (Harper *et al.*, 2010). The invasive procedure involves disruption of cell adhesion and zona pellucida breaching followed by aspiration of one or two blastomeres. The procedure is generally considered safe, though strict evidence of no adverse effect is sparse. The current knowledge is based mainly on a study by Hardy *et al.* (1990), concluding that removal of up to two blastomeres at the 8-cell stage reduced cell numbers in both trophectoderm and inner cell mass of the subsequent blastocysts, but was without adverse effect on development. Later, this conclusion was modified, taking into account that the impact may depend upon the extent of cell loss, the quality of the original embryo and the number of cells present at biopsy (Cohen *et al.*, 2007). This evidence was found in data on embryo cryopreservation, assuming that the cell loss often seen after thawing frozen embryos can be compared with the cell loss after blastomere biopsy. Findings supporting the conclusion have been provided later in studies on fresh transferred embryos. A lower blastocyst rate was found when two cells were removed compared with one cell only (Goossens *et al.*, 2008; De Vos *et al.*, 2009), indicating that the number of removed cells does have an impact on development (Goossens *et al.*, 2008; De Vos *et al.*, 2009). Recent studies on mouse embryos have suggested that blastomere removal results in compaction at an earlier cell stage, and increased frequency of blastocyst contraction and expansion movements (Ugajin *et al.*, 2010). Zona breaching alone, which is a prerequisite for the biopsy, has been

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reported to cause delayed or even abnormal hatching, and factors such as size and number of holes introduced influence the hatching process (Malter and Cohen, 1989; Cohen and Feldberg, 1991; Schmoll *et al.*, 2003; Duncan *et al.*, 2009). A trend towards developmental delay after blastomere biopsy has been observed in both mice and humans (Tarin *et al.*, 1992; Duncan *et al.*, 2009).

The rather unexplored effect of cleavage stage biopsy encourages further investigations of the impact of blastomere removal on preimplantation embryonic development. Recent development and availability of time-lapse incubators designed and approved for clinical use enables continuous monitoring of human embryos in a controlled environment, and safe non-invasive studies of early embryogenesis in humans (Cruz *et al.*, 2011). The present study is the first to evaluate the effect of blastomere biopsy on early human embryonic development using time-lapse analysis.

Materials and Methods

Participants

From August 2010, all couples undergoing PGD treatment at the Centre for Preimplantation Genetic Diagnosis/The Fertility Clinic, Aarhus University Hospital, Denmark were offered participation in the study. Patients consented to time-lapse monitoring and culturing of the embryos in the EmbryoScope™, an incubator with a built-in microscope and camera. The indications for PGD were chromosomal rearrangements and single-gene disorders. Between August 2010 and January 2011, a total of 56 biopsied embryos from 12 PGD couples were included. During the same period, couples admitted to the conventional IVF/ICSI treatment programme were offered blastocyst culturing and time-lapse monitoring as part of a study comparing culture of sibling embryos in the EmbryoScope™ with culture in a conventional incubator (manuscript in preparation). Twenty couples from this study were selected as controls if any of their embryos had 6–8 symmetric blastomeres and <50% of fragmentation when evaluated 68 h after fertilization (i.e. criteria for biopsy in the PGD group). In all 53, 6–8 cell embryos were used as controls. Baseline and cycle characteristics are presented in Table I.

Written informed consent was obtained from each couple before inclusion. The Central Denmark Region Committees on Biomedical Research Ethics approved the study. The project is a part of a larger project registered in Clinical Trial.gov (accession number NCT01139268).

Ovarian stimulation and oocyte retrieval

Patients underwent ovarian stimulation in either a long down regulation protocol (Supracur® Hoechst, Germany) or using a short antagonist protocol (Cetrotide® Serono) using urinary (Menopur®; Ferring Pharmaceuticals A/S, Denmark) or recombinant FSH (Puregon®; MSD) stimulation. FSH doses were adjusted individually according to the patients' ovarian response. A dose of 10.000 IU of hCG (Pregnyl®; MSD) was administered when at least three follicles measured 17 mm or more. Oocyte retrieval was conducted using ultrasound-guided puncture of ovarian follicles 36 h later.

IVF, embryo culture and blastomere biopsy

Following retrieval, oocytes were fertilized using conventional IVF [inseminated for 1.5 h in Sydney IVF fertilization medium (COOK®, Australia)] or ICSI procedures. ICSI fertilized embryos were placed in the EmbryoScope™ immediately after injection, whereas IVF embryos after insemination were moved to 25 μl droplets of Sydney IVF Cleavage Medium (COOK®, Australia) under oil and cultured 18.5 h in a conventional incubator (in both cases at 37°C, 20% O_2, 6% CO_2) before they were transferred for further culturing in the EmbryoScope™. Just prior to transfer to the EmbryoScope™, IVF embryos were denuded using a pipette if cumulus cells were still adhering to ensure optimal image acquisition. In the EmbryoScope™ both IVF and ICSI fertilized embryos were cultured in 25 μl individual droplets of Sydney IVF Cleavage Medium (COOK®) under oil with change to Sydney IVF Blastocyst Medium (COOK®, Australia) 68 h after fertilization. Embryos in the PGD group with 6–8 symmetrical blastomeres, and <50% of fragmentation were selected for biopsy 68–70 h after fertilization. Cell-to-cell adhesions were loosened by short incubation of embryos (2–5 min) in Ca^{2+}- and Mg^{2+}-free bicarbonate buffered medium prior to biopsy (G-PGD® Vitrolife, Sweden). A laser was used to create an ~30 μm opening in the zona pellucida (Zilos™, Hamilton Thorne Research, wavelength 1480 nm), and a single blastomere was aspirated through the hole using an aspiration pipette. After biopsy, during which the embryos were temporarily removed from the time-lapse incubator (but with continued time monitoring), the embryos were returned to the EmbryoScope™ until transfer on Day 5 after oocyte retrieval.

Table I Baseline and cycle characteristics in PGD and control group.

	Control (no biopsy)	PGD (biopsy)	*P*-value
Number of treatment cycles	20	14	
Maternal age (years)	32 (26–37)	35 (26–41)	0.06
Maternal BMI (kg/m^2)	22.3 (19.2–32.1)	23.0 (20.3–30.9)	0.39
Cumulative FSH dose (IU)	1388 (825–2850)	3075 (1000–3950)	0.00
Number of retrieved oocytes/cycle	15.6 ± 5.0	19.4 ± 1.9	0.08
Number of ICSI procedures	12	13	0.05
Number of IVF procedures	8	1	0.05

Where continuous data did not approximate to a normal distribution, data are displayed as medians with range and Wilcoxon's rank-sum test was used (Age, FSH and BMI), where normal continuous data are expressed as mean values ± SD and Student's *t*-test was used (oocytes retrieved). Categorical data are presented as number of cases and differences were tested with Fisher's exact test.

Time-lapse monitoring

The time-lapse system is a commercially available incubator (EmbryoScope™; Unisense Fertilitech, Aarhus, Denmark) with a built-in microscope and camera connected to a computer. Images were recorded automatically in seven planes (15 μm intervals) every 20 min. Registered events and their definitions are listed in Table II. Time-point refers to the point of time, where an image is recorded. The first event registered after biopsy was compaction.

Statistical analysis

Continuous data were analysed as two independent samples from normal distributions based on Student's *t*-test. Assumption of normality was

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Table II Events registered in time-lapse analysis.

Parameter	Description	Definitions
Cell stages	Time-points for each cell division until compaction	A cell division is defined as the first time-point when the two daughter cells are completely separated by a cytoplasmic membrane
Embryonic stages	Time-points and duration of compaction, morula and blastocyst stages	Compaction is the first time-point when blurred boundaries of at least two cells are observed Morula is the time-point when all cells have compacted Early blastocyst is the first time-point when a blastocoel is visible Full blastocyst is the time-point when the blastocoel fills out the embryo Hatching blastocyst is the time-point when zona breaches Hatched blastocyst is the first time-point, when the embryo has fully escaped zona pellucida
Blastocyst contractions	Time-points, duration and extent of blastocyst contractions	Collapse is defined as a time-point, when the measured diameter is smaller than the diameter at the previous time-point Full recovery is defined as the time-point, when the blastocyst diameter is identical to the diameter just before collapse Extent of the collapse is the largest diameter minus the smallest diameter

Table III Time-points of the start of embryonic stages in the PGD and control group.

	Control (no biopsy)		PGD (biopsy)		*P*-value
	n	Time (h) since fertilization	*n*	Time (h) since fertilization	
2-cell stage	53	26.1 (25.2, 26.9)	56	27.3 (26.7, 28.0)	0.02
3-cell stage	53	35.7 (34.3, 37.3)	56	37.3 (36.0, 38.7)	0.13
4-cell stage	53	37.6 (35.9, 39.3)	56	39.2 (37.9, 40.4)	0.14
5-cell stage	53	49.6 (47.3, 52.0)	56	50.3 (46.2, 52.6)	0.68
6-cell stage	53	52.4 (50.1, 54.9)	56	53.8 (51.7, 56.0)	0.40
7-cell stage	51	58.0 (55.5, 60.5)	51	57.3 (55.3, 59.5)[a]	0.69
8-cell stage	45	60.5 (57.8, 63.4)	36	59.5 (57.5, 61.5)[b]	0.54
Compaction	51	79.7 (77.4, 82.0)	49	90.2 (87.8, 92.5)	<0.001
Morula	48	85.4 (83.0, 87.8)	45	95.3 (92.6, 98.1)	<0.001
Early blastocyst	44	95.9 (93.4, 98.5)	43	101.8 (99.1, 104.6)	<0.003
Full blastocyst	41	102.7 (99.9, 105.6)	31	108.4 (105.0, 111.9)	0.01
Hatching start	19	111.9 (107.8, 116.1)	19	109.7 (105.6, 114.0)	0.45

Continuous data are expressed as median values with 95% CI. For testing the hypothesis of no difference between the two groups Student's *t*-test was used after log-transformation of data. The first time-point registered after biopsy was compaction. Therefore, the time-point of the 7-cell stage excludes embryos biopsied at the 6-cell stage (a) and the time-point of the 8-cell stage excludes embryos biopsied at the 6- and 7-cell stage (b). All other time-points include all embryos. The numbers displayed in each group are embryos reaching the recorded stages. Some embryos in the control group initiated compaction after the 6- or 7-cell stage. The difference in embryo numbers between full blastocyst stage and hatching is because some embryos were transferred, vitrified or discarded before they initiated hatching.

checked with histograms and QQ-plots. If data did not fulfil the assumption of normality, the hypothesis of no difference between the groups was tested with the Wilcoxons rank-sum test. For categorical data, Fisher's exact test was used to test the hypothesis of no difference between the two groups. Two-sided *P*-values under 0.05 were considered significant. All statistical analyses were performed in the statistical package STATA, version 11.0 (StataCorp, USA).

Results

Thirty-six embryos were biopsied at the 8-cell stage, 15 embryos at the 7-cell stage and 5 embryos at the 6-cell stage. The duration of the 1-cell stage (corresponding to the time-point of the 2-cell stage) stands out by being shorter in the control group. From the second cell stage until biopsy, no differences were found between the biopsied group and the control group regarding time-points and duration of the cell stages (Tables III and IV). Compared with un-biopsied control embryos, the duration of the cell-stage at which embryos in the PGD group were biopsied lasted significantly longer ($P < 0.001$, Table IV), whereas the duration of the following embryonic stages up to blastocyst was similar in the two groups (Table IV). Accordingly, after biopsy, PGD embryos reached the subsequent embryonic stages until hatching at significantly later time-points ($P_{compaction} < 0.001$; $P_{morula} < 0.001$; $P_{earlyblast} < 0.003$; $P_{fullblast} = 0.01$) (Table III). This

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Table IV Duration of embryonic stages (hours).

Embryonic stage	Control (no biopsy)		PGD (biopsy)		*P*-value
	n	h	*n*	h	
1-cell	53	26.0 (20.5, 34.3)	56	27.3 (23.3, 33.9)	0.01
2-cell	53	11 (1.6, 14.3)	56	11.5 (0, 13.7)	0.23
3-cell	53	1.2 (0, 15.7)	56	0.9 (0, 14.9)	0.35
4-cell	53	14.2 (0, 24.6)	56	13.7 (0, 23.7)	0.92
5-cell	53	1.5 (0, 22)	56	1.8 (0, 26.4)	0.60
6-cell	53	3.3 (0, 30.8)	2[c]	27.8 (27.8, 27.8)[a]	–[c]
			51	2.3 (0, 27.8)[b]	0.20
7-cell	51	2.3 (0, 37.7)	13	31.6 (2.3, 46.5)[a]	<0.001
			36	2.4 (0, 46.5)[b]	0.13
8-cell	43	19.3 (1.3, 36.8)	33	29.55 (3.3, 49.5)[a]	<0.001
Compaction	48	5.1 (1, 15.2)	49	4.5 (0.9, 18.7)	0.39
Morula	44	9.0 (1.7, 32.8)	43	7.7 (0.4, 19.4)	0.07
Early blastocyst	41	6.4 (1, 26.7)	31	7.6 (1, 34.3)	0.31
Blastocyst	19	20.3 (8.6, 38.4)	19	11.8 (3.7, 34.6)	<0.001
Full blastocyst	19	13.1 (7, 22.7)	19	2.5 (0.3, 16)	<0.001

Data are displayed as medians and range. For testing the hypothesis of no difference between the two groups, Wilcoxon rank sum test was used. Durations of the 6-, 7- and 8-cell stages are split to show embryos biopsied at that stage (a) and embryos not yet biopsied (b). The first time-point registered after biopsy was compaction. Therefore, the duration of the 7- and 8-cell stages exclude embryos biopsied previously. All other time-points include all embryos. The numbers displayed in each group are embryos reaching the recorded stages. Some embryos in the control group initiated compaction after the 6-or 7-cell stage. (c) Only two embryos survived biopsy at the 6-cell stage and provide too few for a meaningful statistical analysis. The difference in embryo numbers between duration of the early and the blastocyst/full blastocyst stage is because some were transferred, vitrified or discarded before they initiated hatching. The durations of the blastocyst and full blastocyst stage were calculated as the time elapsed from the beginning of the early and full blastocyst stage, respectively, until hatching and could therefore be obtained for hatching blastocysts only.

was regardless of the number of blastomeres at the time of biopsy (analysis not shown). We found no differences in embryo diameter at the different blastocyst stages (Fig. 1). Nor were any differences in number or duration of contractions detected (Fig. 2). The duration of the complete blastocyst stage, from early blastocyst till hatching, was significantly shorter in the PGD group than among control embryos ($P < 0.001$) due to a shorter time-period from the full blastocyst stage was reached until hatching (Table IV). Thus, embryos in the PGD group started hatching at the same time-point as the control group, but had a significantly smaller size ($P < 0.001$) and a thicker zona pellucida ($P < 0.001$, Fig. 3) at this time. In the control group, time-lapse videos revealed that expansion of the blastocyst caused continuous thinning of zona pellucida until the blastocyst hatched, whereas in the biopsied group no expansion or zona thinning was observed. Instead, the blastocyst hatched through the defect in zona pellucida caused by the biopsy (videos of the different hatching mechanisms in the two groups can be found supplementary).

Out of five embryos biopsied at the 6-cell stage, only two continued development, providing too small a number for a meaningful statistical analysis. To evaluate whether the observed delay was related to subsequent embryonic arrest, nine ($n = 9$) embryos from the control group and thirteen ($n = 13$) embryos from the PGD group not reaching the blastocyst stage were excluded in an additional, but otherwise identical analysis. The results were unchanged; both the observed difference in reaching subsequent time-points combined with the same time-point of hatching ($P_{compaction} < 0.001$; $P_{morula} < 0.001$; $P_{earlyblast} < 0.002$; $P_{fullblast} = 0.01$, $P_{hatching} = 0.45$) and the observed prolonged duration of the biopsed cell stage only, and the shorter duration of the full blastocyst stage in the biopsied group ($P_{7\text{-cell stage}} < 0.001$, $P_{8\text{-cell stage}} < 0.001$, $P_{compaction} = 0.39$; $P_{morula}0.07$; $P_{earlyblast}0.30$; $P_{fullblast} = 0.01$; Table SI).

Discussion

Time-lapse monitoring enables a detailed and dynamic analysis of the development of human pre-implantation embryos. Recording time-points for initiation of embryonic stages permits a thorough comparison of biopsied and non-biopsied embryos. The present study is to our knowledge the first to describe the dynamic development of human embryos after removal of a blastomere using time-lapse analysis.

We found that duration of the cell stage at which biopsy was performed was significantly prolonged in biopsied embryos compared with non-biopsied control embryos, whereas duration of compaction, morula and early blastocyst stage was identical in the two groups. The prolongation of the biopsied cell stage caused PGD embryos to reach the subsequent embryonic stages at later time-points than embryos in the control group. The retarded, but otherwise normal development was present until the beginning of hatching. Biopsied and non-biopsied embryos started hatching at the same time, which caused duration of the blastocyst stage to be shorter in the biopsied group, due to a shorter time from reaching the full blastocyst stage until hatching.

Tarin *et al.* (1992) observed a significantly higher proportion of human embryos reaching the morula stage retarded (after Day 4) following biopsy at Day 2. Similarly, Duncan *et al.* (2009) observed a trend towards developmental delay in mouse embryos that underwent

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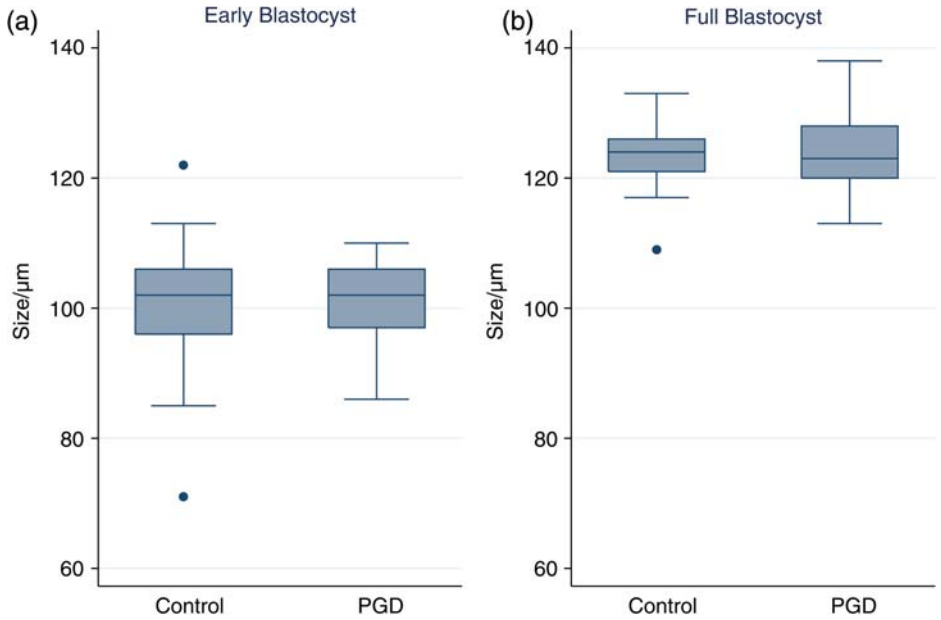


Figure 1 Blastocyst sizes at early (**a**) and full (**b**) blastocyst stage in the non-biopsied control group ($n_{a/b} = 44/41$) and the PGD group ($n_{a/b} = 43/31$). Only embryos reaching the early (a) and full (b) blastocyst stages are included in the figure. The middle band represents the median value, the upper and lower limits of the box represent the upper and lower quartiles, respectively. Whiskers display the upper and lower values within 1.5 times the upper and lower quartiles. Outliers are displayed as dots.

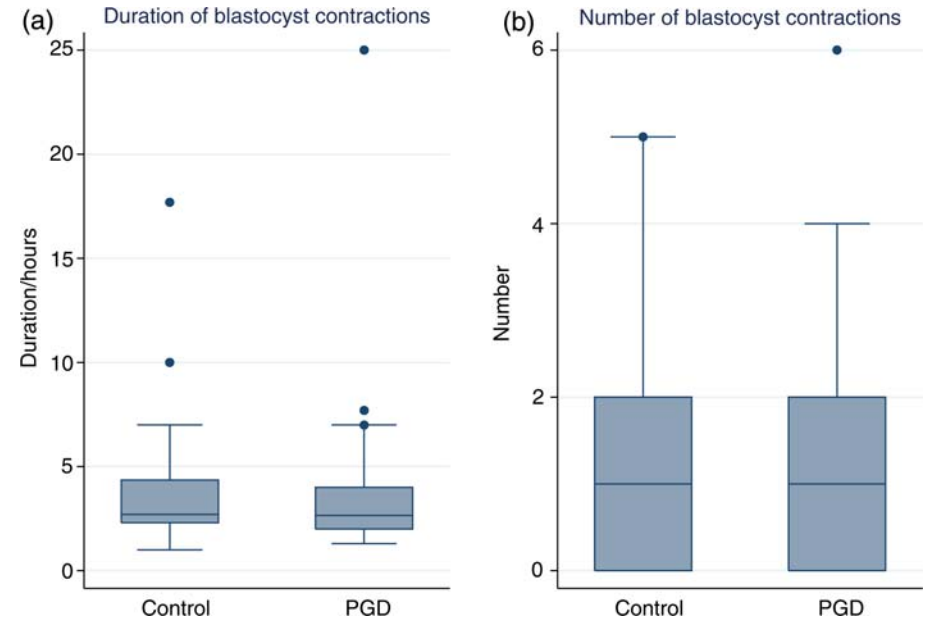


Figure 2 Duration (**a**) and number (**b**) of blastocyst contractions in the non-biopsied control group ($n = 44$) and the PGD group ($n = 43$). Only embryos reaching the early blastocyst stages are included in the figure. The middle band represents the median value, the upper and lower limits of the box represent the upper and lower quartiles, respectively. Whiskers display the upper and lower values within 1.5 times the upper and lower quartiles. Outliers are displayed as dots.

blastomere biopsy compared with untreated controls. Our study confirms these findings and identifies the delay as being caused by a prolongation of the biopsied stage only, resulting in a delayed compaction, after which the embryo continues its normal pattern of development. The morphological appearance of compaction is the flattening of individual blastomeres to maximize contact between neighbouring cells

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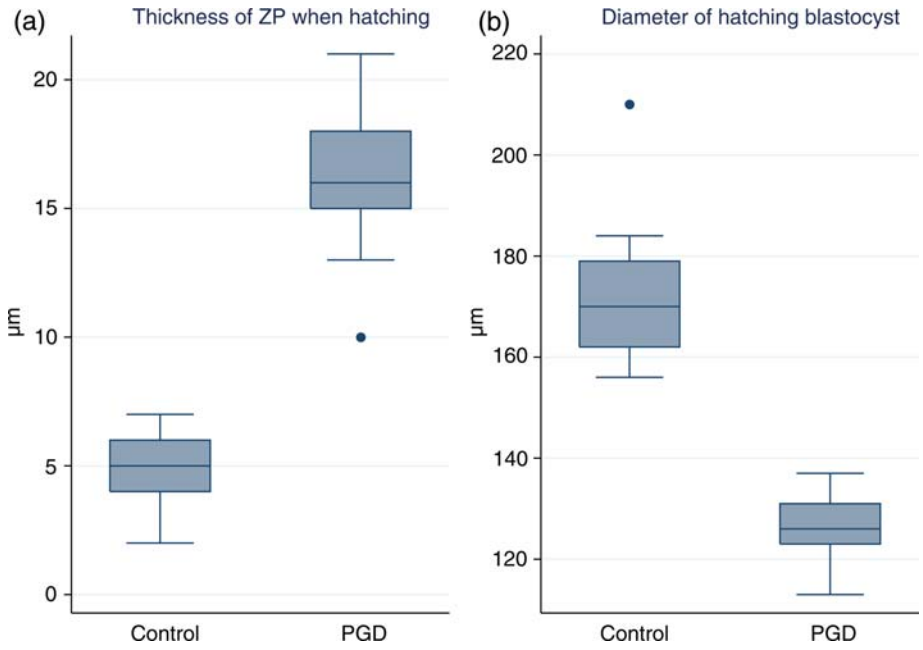


Figure 3 Thickness of zona pellucida (**a**) and diameter of the blastocyst (**b**) when hatching in the bon-biopsied control group ($n = 19$) and the PGD group ($n = 19$). Only embryos reaching the hatching stage are included in the figure. The middle band represents the median value, the upper and lower limits of the box represent the upper and lower quartiles, respectively. Whiskers display the upper and lower values within 1.5 times the upper and lower quartiles. Outliers are displayed with dots.

and increasingly obscured intercellular boundaries. During this process, changes in both inter- and intra-cellular organization occur, including polarization of blastomeres, formation of focal gap junctions, tight junctions and cytoskeletal adherence between blastomeres (Fleming *et al.*, 2000; Fleming *et al.*, 2001). Ca^{2+} plays an important role in many of these changes, probably by maintaining the compacted shape through effects on the cytoskeleton (Pey *et al.*, 1998). In the mouse, Ca^{2+}-depleted medium has been shown to cause re-distribution of the cytoskeletal protein E-Cadherin, abolish normal cell flattening and intercellular apposition, thereby hindering compaction (Pratt *et al.*, 1982; Clayton *et al.*, 1993; Sefton *et al.*, 1996; Pey *et al.*, 1998). In humans, E-Cadherin distribution appears to be stage-dependent with relocation to areas of cell-to-cell contact on Day 4 of development, suggesting a role in compaction (Alikani, 2005). Since PGD embryos were exposed, although briefly, to Ca^{2+}/Mg^{2+}-free medium as part of the biopsy procedure, these known effects would offer a likely explanation for the observed delay in compaction. However, the blastomere removal *per se* may be contributing as well, since a trend towards developmental delay was seen more often in mice embryos subjected to a complete biopsy procedure, than in embryos only subjected to incubation in Ca^{2+}/Mg^{2+}-free medium, zona breach or both (Duncan *et al.*, 2009). Animal studies have shown that the stages just prior to embryonic arrest are significantly prolonged (Holm *et al.*, 1998). We performed an additional analysis, excluding embryos that arrested their development after the biopsy. This exclusion did not change the result, indicating that the overall prolongation in the biopsied cell stage observed was not merely an effect of individual embryos having a prolonged cell stage prior to development arrest.

Tarin *et al.* (1992) observed that biopsy of human embryos at an early stage (Day 2) reduces the number of cells in both the trophectoderm and the inner cell mass, and that the reduction was pronounced mostly in embryos with retarded development (Tarin *et al.*, 1992). Our study addressed the question of embryo volume by measuring the diameter of the embryos. We found that embryo diameter in the two groups was the same at early and full blastocyst stage, respectively. The divergence may reflect differences in biopsy procedure, since embryos in the study by Tarin *et al.* (1992) were biopsied at Day 2 after insemination, where a quarter of the embryos was removed, whereas we biopsied embryos on Day 3 removing only 1/6–1/8 of the embryo. Removal of a quarter of the embryo on Day 2 constitutes a more profound intervention, and the two studies are accordingly not entirely comparable.

Previously, the effect of blastomere removal on the process of hatching has been studied in mice using time-lapse monitoring (Duncan *et al.*, 2009; Ugajin *et al.*, 2010). Studying the effect of biopsy by comparing embryos having one out of four cells removed ($n = 9$) to a non-biopsied control group ($n = 14$), Ugajin *et al.* (2010) observed that the number of contractions was higher and that time to end of hatching was prolonged in the biopsied group. Based on these findings the authors concluded that blastomere removal negatively influenced the hatching process. Duncan *et al.* (2009) observed early and abnormal hatching of mouse embryos. Presently, the precise mechanisms of hatching *in vitro* are not fully explained, but previously a progressive expansion of the blastocyst volume due to increasing hydrostatic pressure within the blastocoele caused by intake of fluid has been observed to cause thinning of zona pellucida (Cohen and Feldberg, 1991; Montag *et al.*, 2000;

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Sathananthan *et al.*, 2003), as observed also in control embryos in the present study. Different phenomena, such as trophectoderm projections (Gonzales *et al.*, 1996), specialized cells named 'zona breakers' (Sathananthan *et al.*, 2003) or localized effects of zona lysine proteases have been suggested to assist hatching (O'Sullivan *et al.*, 2001; Sharma *et al.*, 2006). Hatching *in vivo* appears to include a global lysis of the zona pellucida, suggesting a uterine involvement in *in vivo* hatching consistent with co-expression of lysine proteases in uterine glands (O'Sullivan *et al.*, 2002). Blastocyst contractions *in vitro* have been described in a variety of species, including human (Lewis and Gregory, 1929; Massip and Mulnard, 1980; Massip *et al.*, 1982; Gonzales *et al.*, 1996; Holm *et al.*, 1998; Mio and Maeda, 2008). Mostly these contractions have been regarded as a normal phenomenon, while others have considered the frequent interruption of expansion with repeated contractions to disturb the hatching process (Massip and Mulnard, 1980; Massip *et al.*, 1982). In contrast to the findings by Ugajin *et al.* (2010) in mice, we found no differences between the control and the biopsied group when analysing the number, duration and extent of contractions, and therefore we find no evidence of disturbed process leading to hatching. The disparity of our observations to the previous observations in mice could be explained by the study of mice having more frequent recording (5 min versus 20 min in our study), a smaller population (14 control/9 biopsied versus 53 control/56 biopsied in our study), a larger proportion of cells removed (1/4 versus 1/7 or 1/8 in our study) or it may reflect true differences between species.

The mechanism of hatching was altered after blastomere biopsy in the present study. Embryos in the biopsied group were shown to hatch through a thicker zona pellucida, while having a smaller embryo size. Time-lapse videos of the two groups revealed two different mechanisms of hatching. Embryos in the biopsied group escaped through the artificially induced hole in zona pellucida instead being preceded by the gradual thinning of zona pellucida observed in the continuously expanding embryos in the control group, indicating that the normal mechanism of hatching was circumvented by introducing the relatively large opening in zona pellucida as part of the biopsy procedure. The observations are in complete concordance with previous studies on the mechanism of assisted hatching, where zona-manipulated embryos were shown to hatch through the artificial hole in a thicker zona pellucida, whereas hatching in untreated embryos was preceded by blastocyst expansion and zona thinning (Malter and Cohen, 1989; Montag *et al.*, 2000; Schmoll *et al.*, 2003). As blastomere removal is performed after introducing a breach in the zona pellucida, we consider these known effects of zona manipulation to be the most likely explanation for the altered mechanism of hatching observed in our study. Furthermore, zona manipulation is known to cause earlier hatching, at a time-point that presumably corresponds to the onset of blastocyst expansion, since the pressure caused by the growing blastocyst is relieved through the opening in zona pellucida (Montag *et al.*, 2000). Our study supports the finding that expansion seems unnecessary for hatching in biopsied embryos. Unexpectedly, embryos in the two groups initiated hatching at the same time. The fact that biopsied embryos did not hatch earlier than non-biopsied—despite facilitation of the process by zona breakage—may simply reflect that the delay in the development caused by the prolonged duration of the biopsied cell stage was counterbalanced by early hatching. Thus, embryos in the biopsied group reached the full blastocyst stage at a later time-point, but subsequently spent a shorter time at the full blastocyst stage before hatching. A contributing factor could be reduced embryo volume, since previously biopsied embryos have been shown to consist of fewer cells (Hardy *et al.*, 1990). Further time-lapse studies including a group of embryos with an artificially induced defect in zona pellucida only—and no biopsy—could further elucidate the altered process of hatching after cleavage stage biopsy.

The interpretation of the results is based on an assumption of two comparable groups. All embryos in the two groups were selected by the same senior biologist experienced with PGD. Accordingly, control embryos were selected applying the same criteria for accepting embryos for biopsy in PGD, i.e. 6–8 blastomeres with <50% fragmentation 68 h after fertilization. The control embryos for this study came from the EmbryoScope™ group of a cohort randomised to culture either in the EmbryoScope™ or in a conventional incubator in a safety study comparing the two incubators. The fact that they were randomized just after fertilization should eliminate a potential bias in selection. *Post hoc* time-lapse analysis of development found no differences between the groups regarding time-points and duration of the cell stages from the 2-cell stage until biopsy. The difference found in duration of the 1-cell stage may be explained partly by difference in IVF/ICSI ratio, although the study material is too small to make any definitive conclusions. Early cleavage is an established marker of developmental competence (Shoukir *et al.*, 1997; Sakkas *et al.*, 1998). The difference in early cleavage, with embryos in the control group displaying earlier cleavage than embryos in the PGD group, indicates that embryos in the two groups are not entirely comparable. There are several differences between the two patient groups from which the embryos came. The differences found between the two groups were reflected in different treatment indications, cumulative FSH dose used, and ratio of IVF/ICSI. Patients in the PGD group receive assisted reproductive treatment because they have a single gene or chromosomal disorder. Some are thus potentially fertile and might therefore produce embryos of higher quality. On the other hand, there was a non-significant trend towards a higher maternal age in the PGD group, which is known to be clearly associated with a lower oocyte quality, chromosomal abnormalities and impaired embryo development (Munne *et al.*, 1993; Navot *et al.*, 1994; Janny and Menezo, 1996; Pellestor *et al.*, 2003; Taranissi *et al.*, 2005). The two groups and embryos are therefore not completely comparable, with embryos from the biopsied group possibly having a lower quality. Nevertheless, we consider it plausible that the combination of a very similar pattern of development before biopsy in conjunction with the convincing differences after biopsy reflects an impact of the biopsy. We do not find it plausible that any of the above-mentioned considerations could explain the delay in the embryonic stage after biopsy or the change of the hatching process observed.

In general, previous studies describing the impact of cleavage stage biopsy on human embryos has also been hampered by the difficulty of establishing comparable control groups. Blastocyst biopsy is now increasingly used for obtaining DNA for both PGD and PGS. In future studies, PGD embryos biopsied on Day 5 could constitute such a comparable control group. However, the individual contribution of $Ca^{2+/}Mg^{2+}$ depletion and blastomere removal *per se* to the observed developmental delay would require including a step of $Ca^{2+/}Mg^{2+}$ depletion in the blastocyst biopsy group.

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In summary, the present study demonstrates that although cleavage stage embryo biopsy prolongs the cell stage, at which the biopsy is performed, the cleavage rate after this initial delay seems unaffected. Moreover, the study confirms previous findings from studies on assisted hatching, showing the embryo to hatch through the artificial breach in zona pellucida. The study hereby contributes to the understanding of the effect of blastomere biopsy and the early development of the human embryo.

Supplementary data

Supplementary data are available at http://humrep.oxfordjournals.org/.

Authors' roles

All authors played a role in study conception and design. K.K. and J.H. took part in collection and assembly of data. K.K. carried out data analysis. K.K. and J.I. interpreted the findings and drafted the manuscript. All authors critically reviewed and approved the final version of the manuscript.

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